COMPUTER VISION TUTORIAL

# Understanding GrabCut: A Step-by-Step Guide

A visual, numerical walkthrough of image segmentation using a 3×3 pixel matrix

> **GrabCut Core Principle:** GrabCut segments foreground objects from the background using two complementary sources of information: **(1) Color Statistics** (how close pixel color matches the model) and **(2) Spatial Coherence** (whether neighboring pixels belong to the same group).

## Step 1: Input Image & Initial Classification

Consider a 3×3 grayscale image. The user draws a bounding box indicating that the **bottom-right area** contains the target object:

**Original 3×3 Image (Pixel Intensities)**

| | | |
|---|---|---|
| 10 | 12 | 11 |
| 10 | 80 | 85 |
| 12 | 82 | 90 |

**Initial GrabCut Mask Labels**

| | | |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 3 | 3 |
| 0 | 3 | 3 |

**Label Definitions:** `0 = Definite Background` (outside rectangle) | `3 = Probable Foreground` (inside rectangle)

## Step 2: Build Color Distributions

GrabCut calculates the statistical color profile for each initial region:

**Background Values**

Values: `10, 12, 11, 10, 12`

Mean Intensity: `μ_BG ≈ 11.0`

Learns: Background pixels cluster closely around 11.

**Foreground Values**

Values: `80, 85, 82, 90`

Mean Intensity: `μ_FG ≈ 84.25`

Learns: Foreground pixels cluster closely around 84.

**Note: In production GrabCut, this is modeled with Gaussian Mixture Models (GMMs) with multiple components rather than a single arithmetic average.*

## Step 3: Evaluating an Uncertain Pixel

Suppose an ambiguous pixel with intensity `78` appears in the image. GrabCut compares it against both learned distributions:

```
Distance from Background : |78 − 11| = 67 (Very High Penalty / Unlikely)
Distance from Foreground : |78 − 84| = 6 (Very Low Penalty / Highly Likely)

→ Likelihood Decision: 78 is strongly favored to be FOREGROUND
```

## Step 4: Incorporating Neighboring Context

Color alone can be noisy. GrabCut couples color affinity with spatial neighbor consensus:

| | | |
|---|---|---|
| 10 | 11 | 12 |
| 11 | **78** | 80 |
| 12 | 82 | 85 |

**Pixel 78 surrounded by 80 & 82**

**Color Evidence**
Value 78 matches Foreground mean (84)

+

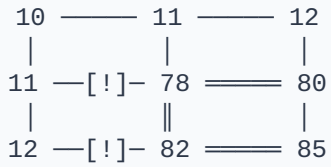
**Spatial Evidence**
Immediate neighbors (80, 82) are Foreground

**Combined Energy Minimization → Confirmed FOREGROUND**

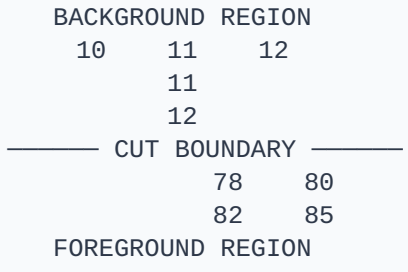
This dual-energy approach is significantly more robust than a naive global intensity threshold (e.g., *"Intensity > 50 = Foreground"*).

## Step 5: Graph Cut Optimization

GrabCut represents the entire pixel grid as a graph with edge weights determining cut penalties:




═══ Strong edge (similar color)
—[!]— Weak edge (large color jump, cuts here)




**Min-Cut separates pixels with minimum energy loss**

## Step 6: Iterative Refinement & RGB Extension

### The Iteration Loop

- Re-assign labels to all uncertain pixels using Min-Cut.
- Update background (10–12) and foreground (78–90) statistics.
- Re-run graph cut with newly tuned energy weights.
- Converges in a few iterations when pixel assignments stabilize.

### Extension to Full RGB Images

In real RGB images, each pixel is a 3D vector:

```
Pixel = [R, G, B] e.g., [120, 80, 40]
```

GMMs fit 3D Gaussian ellipsoids in color space to accurately segment complex textures, skin tones, and soft edges.

### Summary Takeaways

1. **Initialization:** Bounding box seeds hard background and tentative foreground.
2. **Data Term (Color):** Penalizes assigning a pixel to a color model it doesn't match.
3. **Smoothness Term (Contrast):** Penalizes placing segment boundaries across similar neighboring pixels.
4. **Min-Cut** / **Max-Flow:** Finds the mathematically optimal boundary globally and updates iteratively.